

BCA 2nd Semester Exam., 2014

SYSTEM ANALYSIS AND DESIGN

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. **1** and **2** are compulsory.

1. Choose the correct options (any six) : 2×6=12

- (a) How many steps are in the Systems Development Life Cycle (SDLC)?
 - (i) 4
 - (ii) 5
 - (iii) 6
 - (iv) 10
- (b) Which of the following is associated with make-or-buy decision in SDLC?
 - (i) Problem Opportunity
 - (ii) Design
 - (iii) Analysis
 - (iv) Development and Documentation

- (c) In which step actual programming of software code is done in the SDLC?
 - (i) Maintenance and Evaluation
 - (ii) Design
 - (iii) Analysis
 - (iv) Development and Documentation
- (d) Which one is provided by Technical Writers for the new system?
 - (i) Programs
 - (ii) Network
 - (iii) Analysis
 - (iv) Documentation
- (e) Debugging is
 - (i) creating program code
 - (ii) finding and correcting errors in the program code
 - (iii) identifying the task to be computerized
 - (iv) creating the algorithm

(3)

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- (f) Which one of the following is false with respect to a use case?
- (i) It is a scenario for understanding the requirements
 - (ii) It is an interaction between users and the systems
 - (iii) It captures the responsibility of the system to its users
 - (iv) It represents the flow of activities of various processors in system
- (g) Testing of the software without knowledge of source code is called
- (i) white box testing
 - (ii) black box testing
 - (iii) gray box testing
 - (iv) blue box testing
- (h) Which one of the following is the Fact Finding Technique?
- (i) Systems Requirement Specification
 - (ii) Quality Assurance
 - (iii) Sampling of Existing Documents
 - (iv) Prototyping

(4)

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- (i) Traditionally the only phase of software development where a formal approach is used is
- (i) programming
 - (ii) design
 - (iii) planning
 - (iv) testing
- (j) Which one of the following skill may not be required by a Systems Analyst?
- (i) Programming Language Skill
 - (ii) Technical Skill
 - (iii) Communication Skill
 - (iv) Management Skill

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2. Write on any *three* of the following : 4×3=12

- (a) Physical Design
- (b) Management Skill
- (c) Prototyping
- (d) Technical Feasibility
- (e) Real-time Systems

(5)

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Answer any *three* of the following : $12 \times 3 = 36$

3. What is Data Flow Diagram? Explain.
4. Explain different types of Documentation.
5. What are the issues involved in feasibility study?
6. Explain Top-down Design.
7. Discuss the Criteria for Form Design.

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BCA 2nd SEMESTER EXAM., 2015
SYSTEM ANALYSIS AND DESIGN CODE- 303203

Time: 3 hours

Full Marks: 60

Instructions:

- i. The Marks are indicated in the right -hand margin.
- ii. There are **SEVEN** questions in this paper.
- iii. Attempts **FIVE** question in all.
- iv. Question Nos. **1** and **2** are compulsory.

1. Define the following terms (any six). 2*6=12

- | | | |
|--------------------|-------------------------|----------------|
| (a) System | (b) interpersonal Skill | (c) Coupling |
| (d) Decision tree | (e) Audit trail | (f) Testing |
| (g) Questionnaire | (h) Test design | (i) ER diagram |
| (j) Logical design | | |

2. Answer any three of the following: 4*3=12

- (a) Explain the role of the system analyst.
- (b) What do you mean by structured and unstructured interview?
- (c) What do you mean by system requirement specification?
- (d) Explain the role of case tools in system analysis and design.
- (e) Explain the need of data dictionary.

Answer any three of the following: 12*3=36

3. Why are system analysis and design needed? Explain the different types of system.

4. Explain the activities involved in system Development Life Cycle.

5. What is cohesion? Explain the different types of cohesion in detail.

6. Why is system feasibility study essential in system analysis and design? Discuss the various types of feasibility study.

7. What do you mean by form design? What are the different types of form? What are the guidelines that the systems analysts need to follow to design a good form? Explain each of them.

* * *

BCA 2nd Semester Exam., 2016**SYSTEM ANALYSIS AND DESIGN**

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. **1** and **2** are compulsory.

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GROUP—A

1. Define the following in terms of system analysis and design (any six) : $2 \times 6 = 12$

- (a) Distributed system
- (b) Audit trail
- (c) Cohesion
- (d) Data dictionary
- (e) Entity
- (f) Reports
- (g) Prototype
- (h) System design
- (i) DFD akubihar.com
- (j) Quality of software

GROUP—B

2. Answer any *three* of the following : $4 \times 3 = 12$

- (a) 'Interviewing is an art'. Why?
- (b) What do you mean by top-down and bottom-up designs?
- (c) Explain different types of documentation.
- (d) What is joint application design? Explain. akubihar.com
- (e) Explain the essential roles of the system analyst.

GROUP—C

- Answer any *three* questions : $12 \times 3 = 36$

- 3. What is system? What are different elements present in a system? What are different characteristics of a system? Explain.
- 4. What is coupling? Explain different types of coupling in detail with suitable diagram.
- 5. Identify and describe the stages of SDLC.

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6. What is feasibility study? Explain different types of feasibility study. "The objectives of the feasibility study is not to solve the problem but acquire a sense of its scope." Justify your answer.

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7. What is structured analysis? List and explain different types of structured analysis tools.

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Code : 303203

BCA 2nd Semester Theory Examination, 2017

System Analysis and Design

Time : 3 hours

Full Marks : 60

Instructions :

(i) *There are seven questions in this Paper. All questions carry equal marks.*

(ii) *Attempt five questions in all.*

(iii) *Question No. 1 & 2 is compulsory.*

1. (k) Define system.

(l) Define open system.

(m) Define system analyst.

(n) Define design.

(o) Define interaction.

(p) Define abstract system.

(q) What do you mean by investigation?

(r) What is project?

(s) Define file.

(t) Define decision table.

2. Answer any three short type of answer type questions.

(f) List the various characteristics of system.

(g) Distinguish between open and closed systems.

P.T.O.

(h) How important is the informal information system in system analysis?

(i) Describe the role and task of system analyst.

(j) Distinguish between initial investigation and feasibility study.

3. Describe what fact finding techniques would use for investigating the information requirement of a large organization.

4. Discuss the utility of Decision table and Structure chart.

5. Describe the process of design. Also discuss logical and physical design.

6. What is implementation? How does it differ from conversion? Elaborate.

7. List and briefly describe the factors that affect the quality of a system.

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BCA 2nd Semester Exam., 2018

SYSTEM ANALYSIS AND DESIGN

Time : 3 hours

Full Marks : 60

Instructions :

- (i) All questions carry equal marks.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 & 2 are compulsory.

1. Answer any six of the following :

- (a) Define system.
- (b) Define closed system.
- (c) Define physical system.
- (d) What do you mean by static system model?
- (e) Define MIS.
- (f) Define DSS.
- (g) What do you mean by initial investigation?
- (h) Define system planning.
- (i) Define structure chart.

(Turn Over)

8AK/503

(2)

2. Answer any three short type of questions :

- (a) Briefly explain the concept of prototyping.
- (b) Distinguish between interaction and interdependence.
- (c) What is system development life cycle? How does it relate to system analysis?
- (d) What do you mean by strategic and operational planning?
- (e) Distinguish between validity and reliability.

3. Discuss in detail top-down approach to system planning. <https://www.akubihar.com>

4. What are the dataflow diagrams? How are they different from structure charts?

5. Discuss the behavioural issues involved in understanding the analyst/user interface.

6. Under what circumstances would the analysis depend more heavily on external than internal information? Why?

7. Define the term quality assurance. Discuss its importance in system design.

8AK—520/503

Code : 303203

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(2)

BCA 2nd Semester Exam., 2019

SYSTEM ANALYSIS AND DESIGN

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 & 2 are compulsory.

1. Choose the correct answer of the following
(any six) : 2×6=12

- (a) What is black box testing?
- (b) What is exhaustive testing?
- (c) What is static analysis?
- (d) Explain the design steps of the transform mapping.
- (e) How is effort measured?

(Turn Over)

AK9/774

- (f) State the objectives and guidelines for debugging.
- (g) Draw a block diagram of software life cycle model.
- (h) What is software maintenance and categories of maintenance?
- (i) What is software risk?

2. Answer any three of the following : 4×3=

- (a) Describe the characteristics of software.
- (b) Explain the software requirement analysis and modeling.
- (c) Narrate the importance of software specification of requirements.
- (d) Discuss the different levels of testing.
- (e) Write on spiral model in software engineering. <https://www.akubihar.com>

Answer any three of the following questions : 12×3=

- 3. What is feasibility study? Write the types of feasibility. Explain feasibility study process.
- 4. What is structural testing? Explain any two techniques used in it.

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(Continued)

(3)

5. Compare the incremental model and the spiral model.
6. Explain the need for software measures and describe various metrics.
7. Explain SDLC in detail.

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Code : 303203

AK9—640/774

BCA 2nd Semester Exam., 2021

SYSTEM ANALYSIS AND DESIGN

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Choose the correct answer (any six) : $2 \times 6 = 12$

- (a) Which of the following activities does not belong to the implementation phase of the SDLC?
 - (i) File conversion
 - (ii) Program testing
 - (iii) User training
 - ☒ (iv) All of the above
- (b) During what phase, the requirement analysis is performed?
 - (i) System design phase
 - (ii) System development phase
 - ☒ (iii) System analysis phase
 - (iv) System investigation phase

- (c) Which of the following is/are the characteristic(s) of a system?
 - (i) Organization
 - (ii) Interaction
 - (iii) Interdependence
 - ☒ (iv) All of the above
- (d) Which of the following is not considered as a tool at the system design phase?
 - (i) Pie chart
 - ☒ (ii) Data-flow diagram
 - (iii) Decision table
 - (iv) Systems flowchart
- (e) The requirements report includes
 - (i) a hierarchy chart showing the top-level modules
 - (ii) a list of alternative solutions considered
 - (iii) a data-flow diagram describing the proposed new system
 - (iv) All of the above
- (f) Difference between decision tables and decision trees is/are
 - (i) value to end user
 - (ii) form of representation
 - (iii) one shows the logic while other shows the process
 - (iv) All of the above

- (g) The structure chart is
- (i) a document of what has to be accomplished
 - (ii) a statement of information processing requirement
 - (iii) a hierarchical partitioning of the program
 - (iv) All of the above
- (h) The main objective of system evaluation is
- (i) to see whether the system met specification
 - (ii) to improve the system based on operational experience for a period
 - (iii) to remove bugs in the programs
 - (iv) to assess the efficiency of the system
- (i) The main objective of feasibility study is
- (i) to assess whether it is possible to meet the requirements specifications
 - (ii) to assess if it is possible to meet the requirements specified subject to constraints of
 - (iii) budget, human resource and hardware
 - (iv) to assist the management in implementing the desired system

- (j) Problem analysis is done
- (i) during system design phase
 - (ii) during systems analysis phase
 - (iii) before system test
 - (iv) All of the above

2. Answer any *three* of the following : $4 \times 3 = 12$

- (a) What is requirements determination?
- (b) What is feasibility study?
- (c) What is systems analysis?
- (d) Explain joint application development (JAD). <https://www.akubihar.com>
- (e) What is a software project scope?

3. Discuss different types of feasibility study. 12

4. Discuss various system models. 12

5. What is documentation? What are the advantages of documentation? 12

6. What is a system analyst? Discuss the roles of system analyst. 12

7. Explain and differentiate between top-down and bottom-up approach. 12

BCA 2nd Semester Exam., 2022

SYSTEM ANALYSIS AND DESIGN

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Choose the correct answer from any six of the following : 2×6=12

- (a) SRS stands for
 - (i) Software Requirement Specification
 - (ii) Software Requirement Solution
 - (iii) System Requirement Specification
 - (iv) None of the above
- (b) Software engineering aims at developing
 - (i) reliable software
 - (ii) cost effective software
 - (iii) reliable and cost effective software
 - (iv) None of the above

- (c) Which of the following is a tool in design phase?
 - (i) Abstraction
 - (ii) Refinement
 - (iii) Information hiding
 - (iv) All of the above
- (d) Design phase includes
 - (i) data, architectural and procedural designs only
 - (ii) architectural, procedural and interface designs only
 - (iii) data, architectural and interface designs only
 - (iv) data, architectural, interface and procedural designs
- (e) In object-oriented design of software, objects have
 - (i) attributes and names only
 - (ii) operations and names only
 - (iii) attributes, names and operations
 - (iv) None of the above
- (f) A graphic representation of an information system is called
 - (i) flowchart
 - (ii) pictogram
 - (iii) data flow diagram
 - (iv) histogram

- (g) Data structure suitable for the application is discussed in
- (i) data design
 - (ii) architectural design
 - (iii) procedural design
 - (iv) interface design
- (h) An important aspect of coding is
- (i) readability
 - (ii) productivity
 - (iii) to use as small memory space as possible
 - (iv) brevity
- (i) Which of the following comments about object-oriented design of software is not true?
- (i) Objects inherit the properties of class
 - (ii) Classes are defined based on the attributes of objects
 - (iii) An object can belong to two classes
 - (iv) Classes are always different
- (j) A good specification should be
- (i) unambiguous
 - (ii) distinctly specific
 - (iii) functional
 - (iv) All of the above

2. Answer any *three* of the following : $4 \times 3 = 12$
- (a) What is software process or software development life cycle (SDLC)?
 - (b) Differentiate between validation and verification.
 - (c) What is feasibility study?
 - (d) What are coupling and cohesion in software engineering?
 - (e) What is a software project scope?
3. What do structured systems analysis and design method mean? Explain. 12
4. Who is a system analyst? What are the roles of system analyst? 12
5. What is process documentation? Why is it needed? 12
6. What is joint application development? Explain its benefits. 12
7. Describe the prototyping process of designing forms and reports. What deliverables are produced from this process? Are these deliverables the same for all types of system projects? Why or why not? 12

BCA 2nd Semester Exam., 2023

SYSTEM ANALYSIS AND DESIGN

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six of the following : $2 \times 6 = 12$

- (a) What is the need of system analyst?
- (b) What is the purpose of real-time system?
- (c) Define distributed system.
- (d) What is the use of project identification?
- (e) What is the basic purpose of physical design?
- (f) What is the difference between management skills and interpersonal skills?

- (g) What is the meaning of 'quality of software'?
- (h) What is the basic idea of modularity?
- (i) What is the basic purpose of ER diagram?
- (j) What is the criterion of report design?

2. Answer any three of the following : $4 \times 3 = 12$

- (a) How to define forms and what is the importance of forms?
- (b) How is agile approach different from SDLC?
- (c) Explain the difference between logical design and physical design.
- (d) How do coupling and cohesion improve modular and structured designs?
- (e) Explain the role and duties of system analyst.

3. Discuss prototyping and participatory design. 12

4. Describe the difference between system requirement specification and system design specification. 12

5. With respect to modular and structured designs, describe the difference between top-down design and bottom-up design approaches. 12
6. With respect to process of system planning, discuss the issues involved in feasibility study, technical feasibility and operational feasibility. 12
7. Discuss decision tree and decision tables with example. 12

Code : 303203

BCA 2nd Semester Exam., 2025

SYSTEM ANALYSIS AND DESIGN

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.*
- (ii) There are **SEVEN** questions in this paper.*
- (iii) Attempt **FIVE** questions in all.*
- (iv) Question Nos. **1** and **2** are compulsory.*

1. Choose the correct answer (any six) : 2×6=12

(a) Cost-benefit analysis is performed during

- (i) analysis phase**
- (ii) design phase**
- (iii) feasibility study phase**
- (iv) implementation phase**

(b) The advantage of system flowcharts is

- (i) effective communication**
- (ii) effective analysis**
- (iii) queasier group or relationships**
- (iv) All of the above**

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(Turn Over)

- (c) _____ is a tabular method for describing the logic of the decisions to be taken.
- (i) Decision table
 - (ii) Decision tree
 - (iii) Decision method
 - (iv) Decision data
- (d) A DFD is normally levelled as
- (i) it is a good idea in design
 - (ii) it is recommended by many experts
 - (iii) it is easy to do it
 - (iv) it is easier to read and understand a number of smaller DFDs than one large DFD
- (e) System development process is also called as
- (i) system development life cycle
 - (ii) system life cycle
 - (iii) Both (i) and (ii)
 - (iv) system process cycle
- (f) SDLC stands for
- (i) System Development Life Cycle
 - (ii) Structure Design Life Cycle
 - (iii) System Design Life Cycle
 - (iv) Structure Development Life Cycle

- (g) In a DFD, external entities are represented by a/an
- (i) rectangle
 - (ii) ellipse
 - (iii) diamond-shaped box
 - (iv) circle
- (h) The first step of the system analysis phase of the SDLC is to
- (i) propose changes
 - (ii) analyze data
 - (iii) gather data
 - (iv) write system analysis report
- (i) In a feasibility study, the _____ feasibility includes a study of organizational behaviour.
- (i) economic
 - (ii) technical
 - (iii) behavioural
 - (iv) logical
- (j) The final step of the implementation phase of the SDLC is to
- (i) train the users
 - (ii) develop documentation
 - (iii) select the conversion type
 - (iv) write the implementation report

2. Answer any *three* of the following : 4×3=12

- (a) Write short notes on coupling and cohesion.
- (b) List the functions of system analyst.
- (c) Differentiate verification and validation.
- (d) What is a decision tree?
- (e) What is a decision table?

3. Explain the phases of SDLC. 12

4. Discuss process documentation. 12

5. What is a feasibility study? Explain. 12

6. Discuss software process model. 12

7. Describe the prototyping process of designing forms and reports in detail. 12

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